

Student Performance related to video-watching behavior

Github Link: <https://github.com/ECEDataScience/miniproject-s24-CamiCalderong.git>

Description of Data Set Worked

The Path chosen was 'Student Performance related to video-watching behavior', and with the given data file 'behavior_performance.txt' it was possible to analyze some of the relationships between the variables given.

This set is related to data for an online course on how students watched videos and how they performed on in-video quizzes. This data set contains a total of ten columns, from which eight were used for analysis. These variables are:

- fracSpent: is the fraction of time the student spent watching the video (either playing or paused), relative to the length of the video.
- fracComp: is the fraction of the video the student watched, between $\text{fracComp}=0$ (none) and $\text{fracComp} \geq 0.9$ (completely).
- fracPaused: is the fraction of time the student spent paused on the video, relative to the length of the video. So if someone was paused for 0.5 min on a 3 minute video, this would be $\text{fracPaused}=1/6$
- numPauses: is the number of times the student paused the video.
- avgPBR: is the average playback rate that the student used while watching the video, ranging between 0.5x and 2.0x.
- numRWs: is the number of times the student skipped backwards (rewind) in the video.
- numFFs: is the number of times the student skipped forward (fast forward) in the video.
- s: is whether the student was correct ($s=1$) or not ($s=0$) on their first attempt at answering the question given directly after the video.
- fracPlayed: is the fraction of the video watched by the student.

Analysis and Questions

For the first part of the code the data had to be processed in order to eliminate the columns containing the ID and other information that was not important for the analysis. Then, it

was necessary to use a Python built-in function in order to handle the missing values, and those students that did not meet the requirements for the analysis. After that, the data got scaled and transformed into a new data set that is later going to be used for the analysis.

In order to resolve the first question, the k-means clustering method was utilized. The data was divided into three clusters (further explained in the results section), and the data set is modified once again based on this. Further in the analysis, two classes are created and a new column in the data set is built. This new column labels the s variable as 'Correct' or 'Incorrect'.

For the second question, linear regression was used as the method in order to see if a student's watching behavior can be used to predict the performance on the quiz. The first step was splitting the data into x and y variables, and applying the regression method for the s column. Then, this data was divided into a training and a testing set, then the linear model was created with the training data. Finally, this regression model was used to predict the testing data and calculate the mean squared error of the data.

For the third question, the data was grouped into four categories in order to plot and create a more specific classification report. These categories were numPauses and fracPaused, fracComp and fracPlayed, avgPBR and stdPBR, numRWs and numFFs.

In order to carry this last analysis a kNN classification method was used. The K-nearest neighbor method was used because it does not require any previous or current assumptions about the data. It is also a simple concept. For this analysis, the data had to be split into testing and training sets, and scaled. Then the k neighbors model function was applied and fitted and the MSE value was produced.

For further analysis on this question, a confusion matrix was created for each of the categories. This included variables such as precision, recall, the f-1 score and support, also the accuracy of the model used for this data set. The accuracy explains the fraction of correct predictions made by the used model, the precision explains the fraction of correct predictions in the predicted positive class and the recall explains the fraction of correct predictions in the actual positive class. The f1-score explains the relationship between precision and recall and the support is the amount of data points analyzed.

Results of each analysis

How well can the students be naturally grouped or clustered by their video-watching behavior?

According to the data given, the students can be grouped in several categories. One of them is time- related category, meaning `fracSpent`, `fracPaused`, `fracComp`, `fracPlayed`; count-related category, for example `numPauses`, `numRWs`, `numFFs`; and performance-related category, like `avgPBR`, `stdPBR` and `s`.

Also, another important analysis metric is `s`, all the analysis done throughout the project is based on the correctness or incorrectness of each students' answers. For example, the data was classified in the `s` value (0 or 1) of the student. This is the output showcasing the first ten students.

Question 1: Cluster Data

Class

- 0 Correct
- 1 Correct
- 2 Incorrect
- 3 Correct
- 4 Incorrect
- 5 Incorrect
- 6 Correct
- 7 Incorrect
- 8 Correct
- 9 Incorrect

Can a student's video-watching behavior be used to predict a student's performance?

In order to answer this question, a linear regression model was used. The result was a mean squared error (MSE) of 1.0134454238348125. In the context of predicting a student's performance using their video-watching behavior, a MSE this high suggests that there is some level of error in the predictions made using this specific model.

Some reasons why this model is performing poorly might have to do with the complexity of the relationship, missing important features of the analysis, and overfitting or underfitting.

The relationship between the watching behavior of the students and their performance on the online quiz includes a complex influence on multiple factors, and linear regression models

might struggle to capture this complex relationship. The features used in the linear regression model might also not capture all relevant information influencing a student's performance. Also, this linear regression model might be overfitting or underfitting the data. This would lead to poor generalization to the data, and high bias and prediction errors.

*How well can you predict a student's performance on a *particular* in-video quiz question based on their video-watching behaviors while watching the corresponding video?*

The MSE for this model was 0.003904853305288389 which indicates better predictive performance from the model used. The classification reports for the kNN model were the following:

Accuracy: 0.9986351228389445

Classification Report [numPauses - fracPaused]:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	3030
1	1.00	1.00	1.00	5762
accuracy		1.00		8792
macro avg	1.00	1.00	1.00	8792
weighted avg	1.00	1.00	1.00	8792

Accuracy: 0.998066424021838

Classification Report [fracComp - fracPlayed]:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	3030
1	1.00	1.00	1.00	5762
accuracy		1.00		8792

macro avg	1.00	1.00	1.00	8792
weighted avg	1.00	1.00	1.00	8792

Accuracy: 0.998066424021838

Classification Report [avgPBR - stdPBR]:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	3030
1	1.00	1.00	1.00	5762
accuracy			1.00	8792
macro avg	1.00	1.00	1.00	8792
weighted avg	1.00	1.00	1.00	8792

Accuracy: 0.9984076433121019

Classification Report [numRWs - numFFs]:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	3030
1	1.00	1.00	1.00	5762
accuracy			1.00	8792
macro avg	1.00	1.00	1.00	8792
weighted avg	1.00	1.00	1.00	8792

These models achieve high accuracy, close to 1, which indicates that it is making the correct predictions for most instances. The precision and recall for both classes (0 and 1) are 1, indicating good precision and recall for both classes. The f-1 score is also high, indicating a perfect balance between precision and recall for all instances.

Overall, these reports indicate that the kNN model performs very well in predicting a student's performance on a particular in-video quiz question based on their watching behaviors. The high accuracy suggests that the model is effective in making accurate predictions across different feature pairs.



